observer model with three parameters captures PSS, sensory noise and an interval bias (i.e., a tendency to select one interval in preference to the other under uncertainty).

The 2xSJ task provides estimates that correlate fairly well with equivalent parameters estimated using TOJs, SJs, and ternary tasks. However, each trial takes longer than in those single-presentation tasks, which makes experiments more onerous. There are a few reasons why the roving-standard 2xSJ is still worth considering. Firstly, it asks about synchrony explicitly (unlike the TOJ) and by requiring relative judgements it reveals a point of maximal synchrony perception (whereas the SJ and ternary tasks often reveal a range of SOA values that are classified as synchronous). Secondly, it can be added in to a single-presentation task (as a follow-up question every two trials), which somewhat mitigates the burden of additional experimental time. Finally, a case can be made that it will be more resistant to some forms of decision-level bias (Morgan, Grant, Melmoth, & Solomon, 2015; Morgan, Melmoth, & Solomon, 2013). As with the other tasks I have described, code to fit data from the 2xSJ accompanies this chapter.[23] For further information, read the comments there and consult Yarrow et al. (2016).

## 12 Conclusion

In this chapter, I have outlined the benefits of fitting formal observer models to judgements about simultaneity, and described how this can be achieved using Matlab code (see book's GitHub repository). In doing so, I have presented one particular observer model in some detail, and highlighted the fundamentally subjective nature of the SJ task, which requires us to think carefully about how both the strategic decisions and perceptual sensitivity of a participant can affect their psychometric function. I have gone on to supply a brief overview of appropriate models for several closely related timing tasks. I hope I have also provided enough of a tutorial regarding bespoke model fitting and evaluation to allow the interested reader to go forward and explore their own models of perceived simultaneity. Modelling may seem intimidating, but in fact, a good understanding of just a few basic concepts (which is best gained through practical exploration) will take you a long way, providing tools to engage more fully with the timing literature. This is an endeavour I would very much encourage!

23 <TwoAFCSimultaneity_3PEq_Multistart_rawdata>.